

Introduction to DEVOPS (Merged - CSIZG514/SEZG514)

(S1-25) Assignment - 2

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GitHub Repo Link : <https://github.com/PLS-PHANI-TEJA-2024tm93573/aceest-web-devops>

Assingment 2

Please find the relevant links and screen shots present in this document (Please ZOOM for clear images)

Release notes present in tags section in github

Github repository :

<https://github.com/PLS-PHANI-TEJA-2024tm93573/aceest-web-devops>

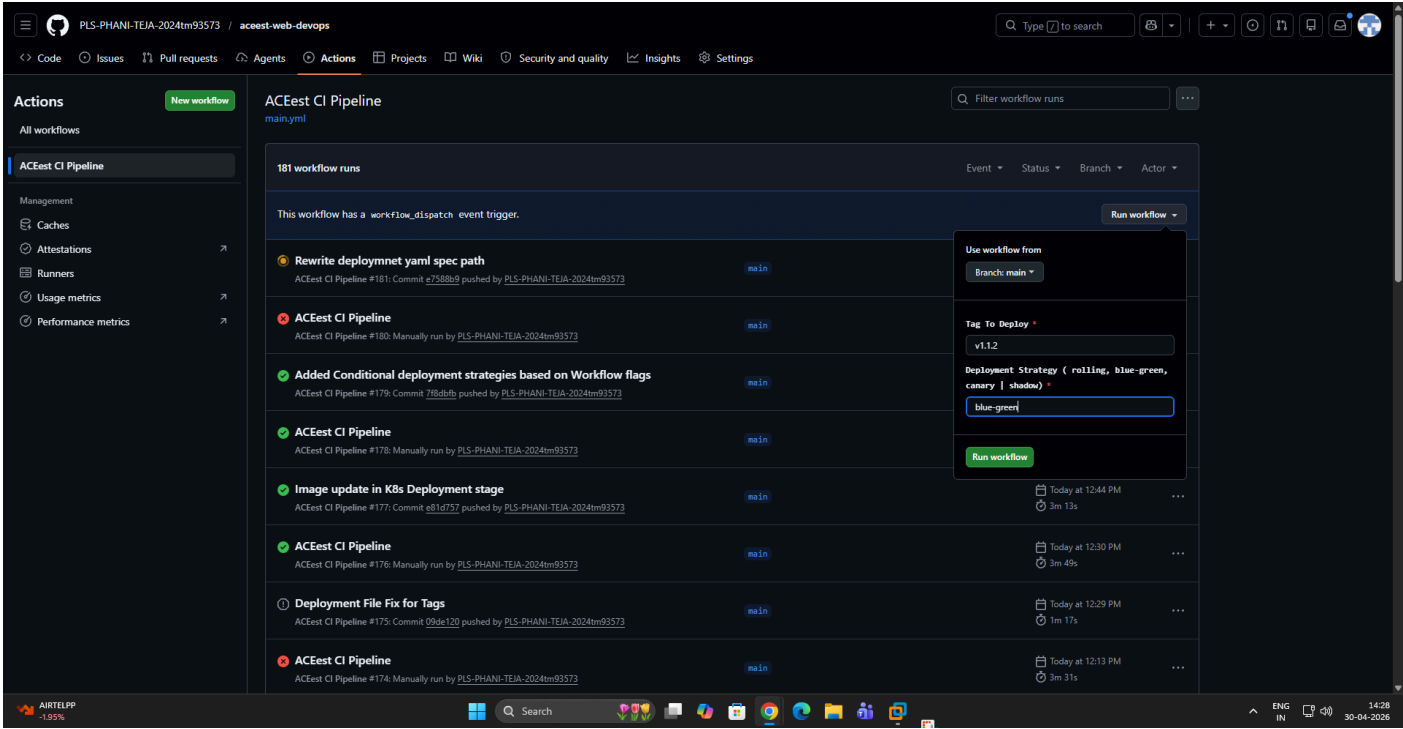
Docker Hub Repository:

<https://hub.docker.com/r/plsphaniteja2024tm93573/aceest-web-app-2024tm93573/tags>

- The Jenkins is triggered by Github Pipeline and given to the github runner running in a linux machine locally .
- Sonar qube works in the same way.
- Deployment is carried out in a kubernetes cluster hosted using **minikube** in a linux machine running locally.

Manual tag Deployment Workflow.

- Each tag is pushed and built and its image is pushed into docker repository.
- To deploy tags into production (Minikube) , a manual workload_dispatch event is used to manually trigger the pipeline to deploy any available tag on demand.
- Rollback Strategy : This gives more control to the engineer to move to any version on demand.
- This is specified in the [github workflow file](#).

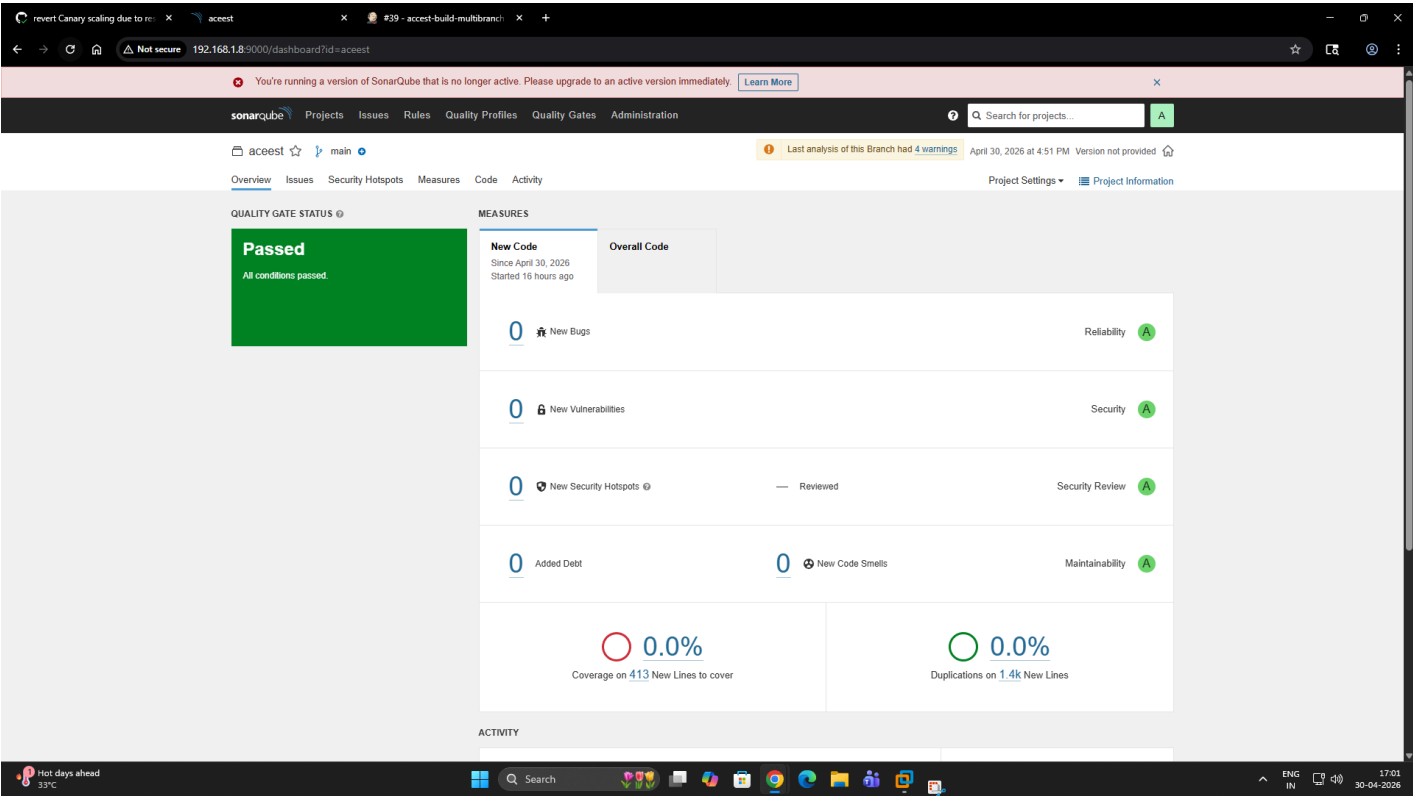
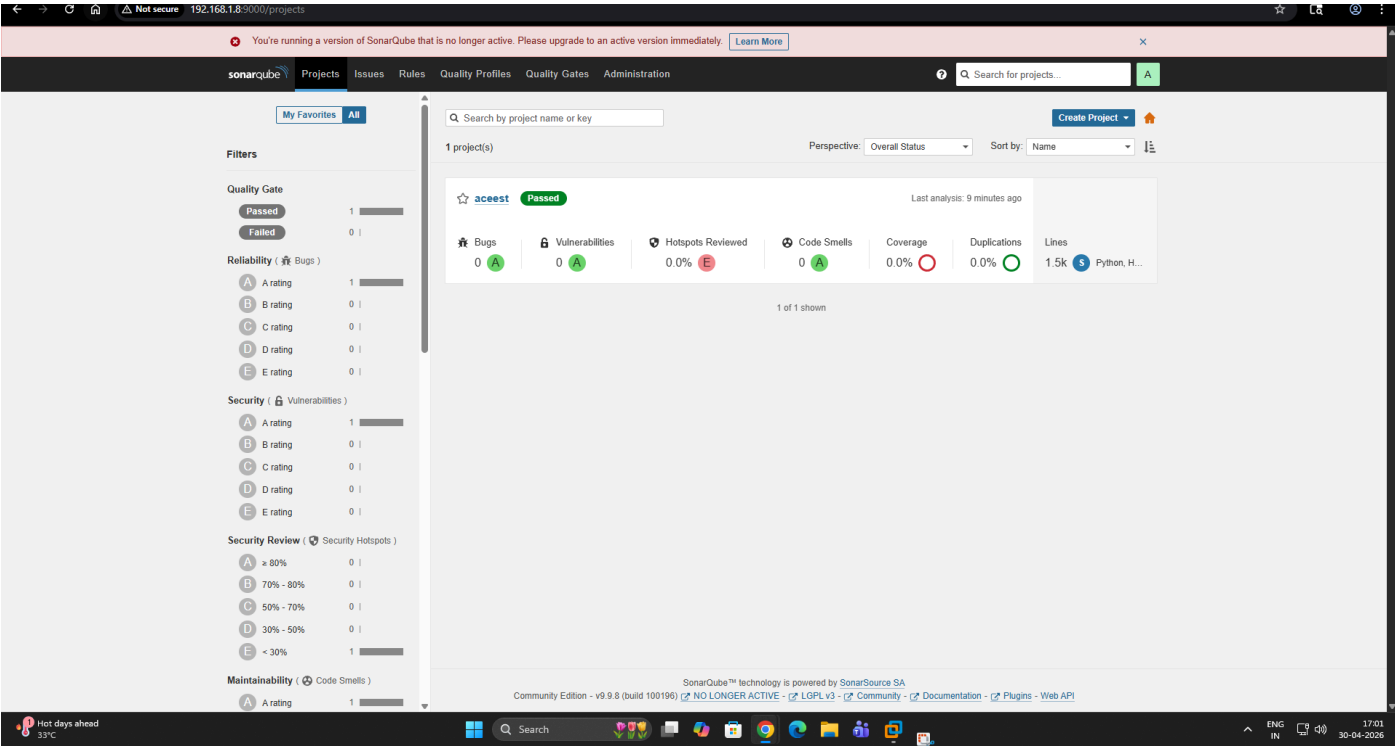


CI - CD Architecture OverView

CI/CD architecture overview is included in the Repository mentioned above. Please refer this link

[CI/CD Architecture Overview](#)

SonarQube Results



Jenkins Pipeline

revert Canary scaling due to re: Issues

#39 - aceest-build-multibranch

192.168.1.8:8081/job/aceest-build-multibranch/job/main/39/stages/

Jenkins

aceest-build-multibranch

main

#39

Stages

Started by admin

Started 3 hr 7 min ago

Queued 6 sec

Took 33 sec

Graph

Start

Checkout SCM

Checkout Repository

Create Python...

Build Validation

Build Docker Image

Build Docker Test...

Run Tests in Docker...

Push Docker Image

End

Search

Push Docker Image

5s

Started 3h 7m ago

Jenkins

...

Checkout SCM 2s

Checkout Repository 0.77s

Create Python Environment 4s

Build Validation 0.29s

Build Docker Image 1s

Build Docker Test Image 0.79s

Run Tests in Docker Container 11s

Push Docker Image 5s

echo "Pushing Docker image as dev-latest..."

docker tag aceest-web-devops:latest plspaniteja2024tm93573/aceest-web-app-2024tm93573:dev-latest

docker push plspaniteja2024tm93573/aceest-web-app-2024tm93573:dev-latest

The push refers to repository [docker.io/plspaniteja2024tm93573/aceest-web-app-2024tm93573]

c97b0504e566: Waiting

858b34b74ec9: Waiting

c7a4c5210d8e: Waiting

3531af2bc2a9: Waiting

4d873bcef452: Waiting

3fc9d9ab5045: Waiting

97a0e483eb8f: Waiting

cb8c0a9140ac: Waiting

5da1b9653b94: Waiting

5da1b9653b94: Waiting

Hot days ahead

33°C

Search

17:00

30-04-2026

revert Canary scaling due to re: Issues

main [aceest-build-multibranch]

192.168.1.8:8081/job/aceest-build-multibranch/job/main/

Jenkins

aceest-build-multibranch

main

Status

Changes

Build Now

View Configuration

Stages

Pipeline Syntax

main

Full project name: aceest-build-multibranch/main

Permalinks

Last build (#39), 1 min 1 sec ago

Last stable build (#39), 1 min 1 sec ago

Last successful build (#39), 1 min 1 sec ago

Last failed build (#29), 1 day 19 hr ago

Last unsuccessful build (#29), 1 day 19 hr ago

Last completed build (#39), 1 min 1 sec ago

Builds

Filter

Today

#39 13:52

#38 12:46

#37 12:32

#36 12:15

#35 12:11

#34 12:05

#33 11:57

#32 01:27

29 April 2026

#31 20:49

28 April 2026

#30 18:46

#29 18:39

#28 18:37

#27 18:01

Light rain

At night

Search

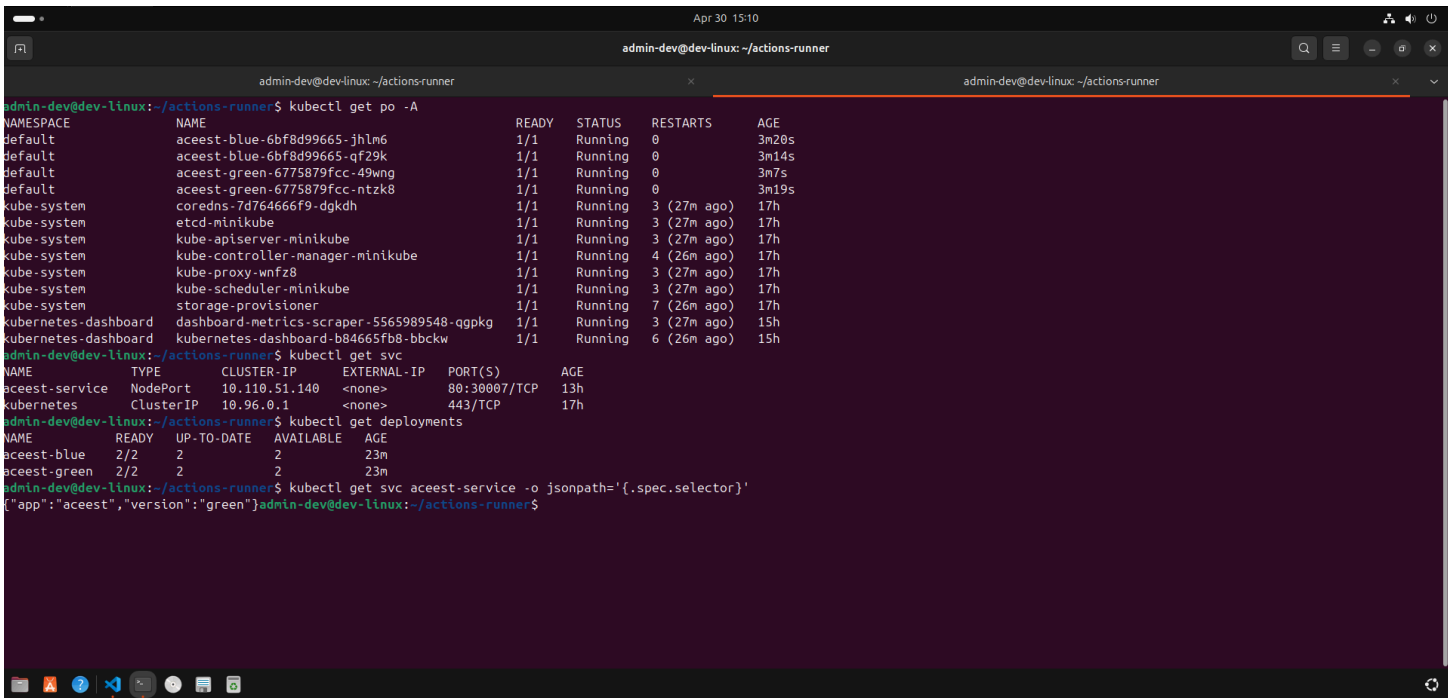
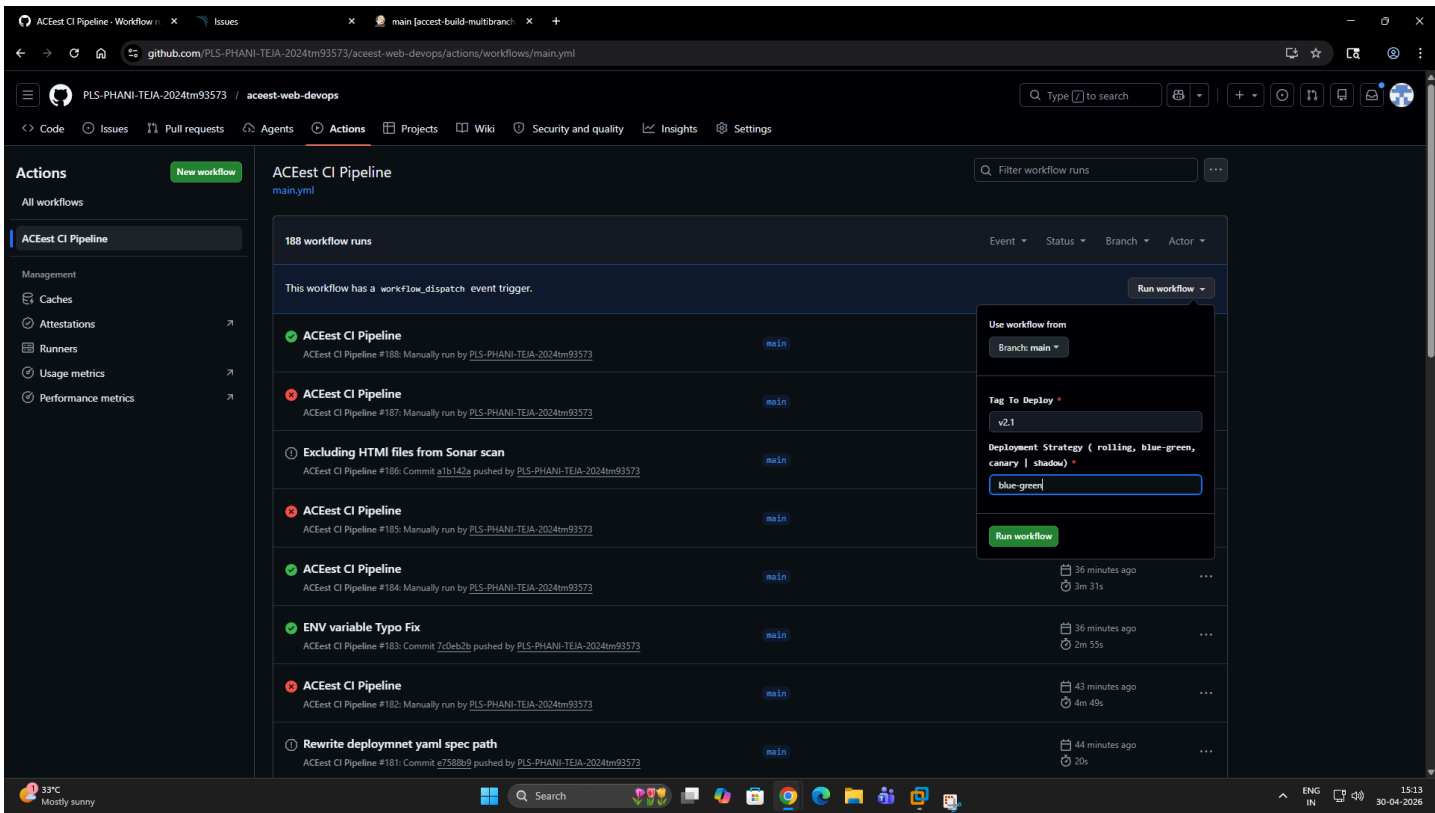
16:59

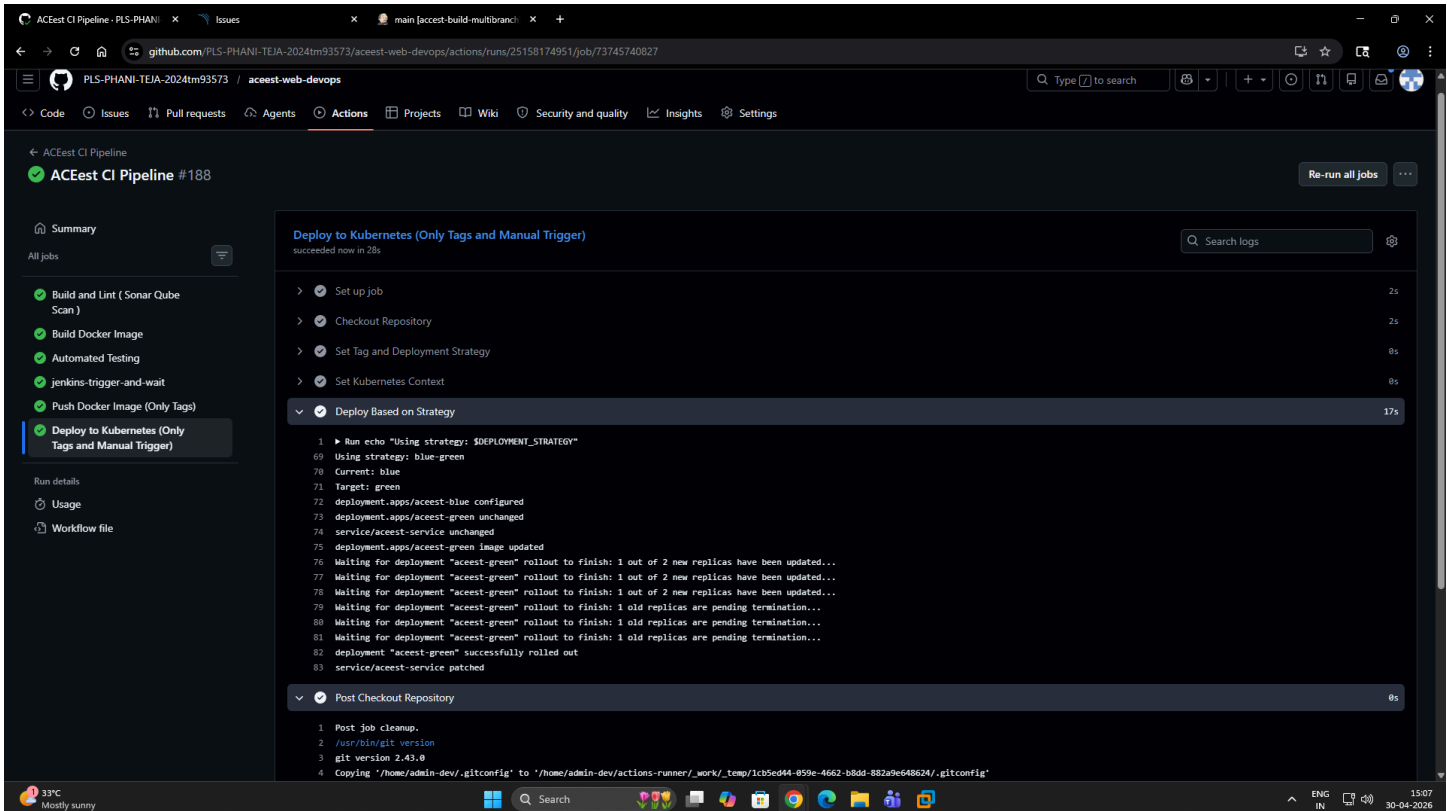
30-04-2026

Deployment Strategies

1. Blue Green Strategy

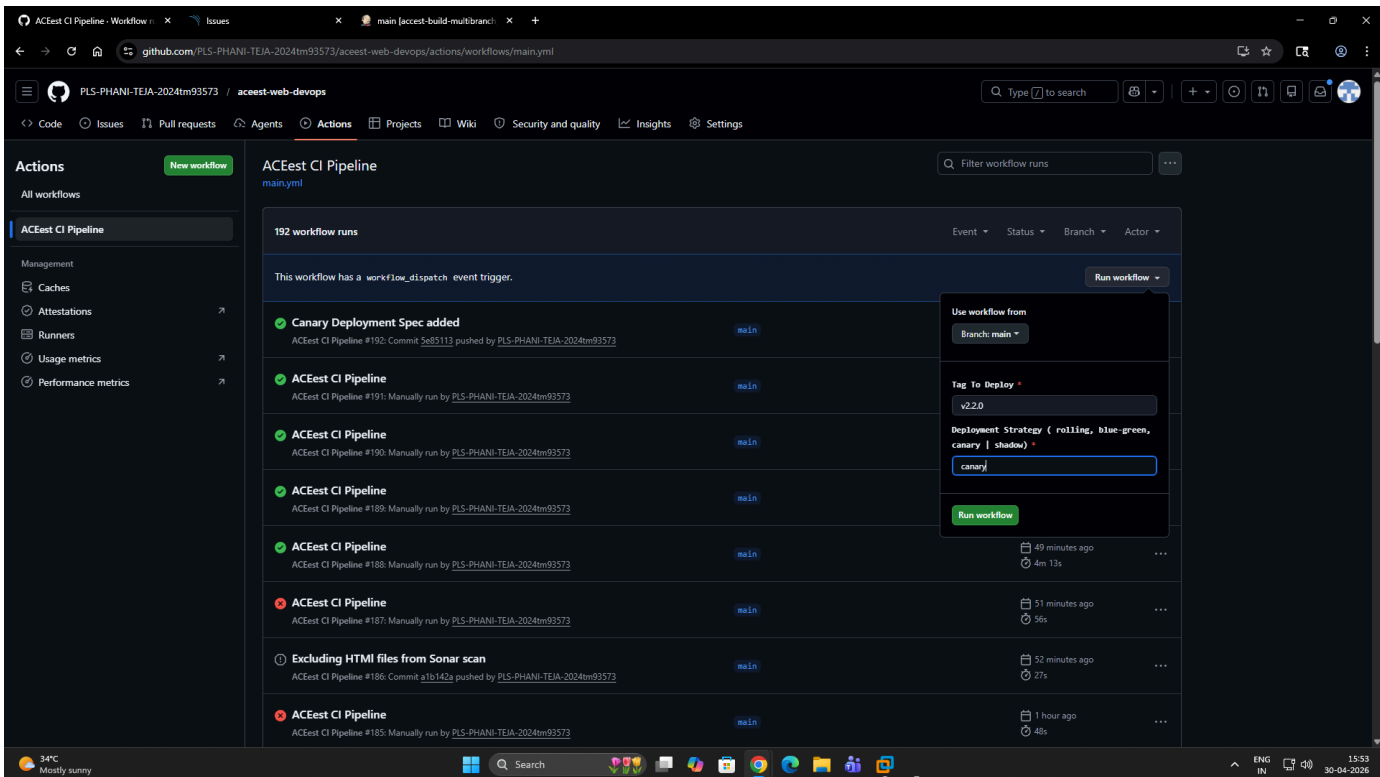
a. [Deployment Spec files \(Github Repository \)](#)





2. CANARY DEPLOYMENT

a. [Deployment Spec Files \(Github Repositories \)](#)



```
admin-dev@dev-linux: ~/actions-runner

admin-dev@dev-linux:~/actions-runner$ kubectl get po -A
NAMESPACE      NAME                                     READY   STATUS    RESTARTS   AGE
default        aceest-app-58b4c85d67-hxxk2           1/1     Running   0           21m
default        aceest-app-58b4c85d67-z447k           1/1     Running   0           21m
default        aceest-canary-6766bcd7f7-mx22j        1/1     Running   0           3m20s
kube-system    coredns-7d764666f9-dgkdh             1/1     Running   3 (82m ago)  18h
kube-system    etcd-minikube                         1/1     Running   3 (82m ago)  18h
kube-system    kube-apiserver-minikube               1/1     Running   3 (82m ago)  18h
kube-system    kube-controller-manager-minikube      1/1     Running   4 (81m ago)  18h
kube-system    kube-proxy-wnfz8                      1/1     Running   3 (82m ago)  18h
kube-system    kube-scheduler-minikube               1/1     Running   3 (82m ago)  18h
kube-system    storage-provisioner                   1/1     Running   7 (81m ago)  18h
kubernetes-dashboard  dashboard-metrics-scraper-5565989548-qgpgk  1/1     Running   3 (82m ago)  16h
kubernetes-dashboard  kubernetes-dashboard-b84665fb8-bbckw      1/1     Running   6 (81m ago)  16h

admin-dev@dev-linux:~/actions-runner$ kubectl get svc
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
aceest-service      NodePort    10.110.51.140 <none>         80:30007/TCP    14h
kubernetes           ClusterIP   10.96.0.1     <none>         443/TCP          18h

admin-dev@dev-linux:~/actions-runner$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
aceest-app          2/2     2             2           21m
aceest-canary       1/1     1             1           3m36s

admin-dev@dev-linux:~/actions-runner$ kubectl get pods -l track=canary -o=jsonpath='{.items[*].spec.containers[*].image}'
plspaniteja2024tm93573/aceest-web-app-2024tm93573:v2.2.0admin-dev@dev-linux:~/actions-runner$
```

ACTest CI Pipeline - PLS-PHANI

Issues

main [accept-build-multibranch]

+

github.com/PLS-PHANI-TEJA-2024tm93573/aceest-web-devops/actions/runs/25160457032/job/73753655014

Summary

All jobs

Build and Lint (Sonar Qube Scan)

Build Docker Image

Automated Testing

jenkins-trigger-and-wait

Push Docker Image (Only Tags)

Deploy to Kubernetes (Only Tags and Manual Trigger)

Run details

Usage

Workflow file

Deploy to Kubernetes (Only Tags and Manual Trigger)

succeeded now in 30s

Set Tag and Deployment Strategy

Set Kubernetes Context

Deploy Based on Strategy

Post Checkout Repository

Complete job

1 ▶ Run kubectl config use-context minikube

7

7 Switched to context "minikube".

1 ▶ Run echo "Using strategy: \$DEPLOYMENT_STRATEGY"

69 Using strategy: canary

70 Canary Deployment

71 deployment.apps/aceest-canary created

72 service/aceest-service configured

73 deployment.apps/aceest-canary image updated

74 Waiting for deployment "aceest-canary" rollout to finish: 1 old replicas are pending termination...

75 Waiting for deployment "aceest-canary" rollout to finish: 1 old replicas are pending termination...

76 Waiting for deployment "aceest-canary" rollout to finish: 1 old replicas are pending termination...

77 deployment "aceest-canary" successfully rolled out

1 Post job cleanup.

2 /usr/bin/git version

3 git version 2.43.0

4 Copying /home/admin-dev/actions-runner/work/.tmp/d8e75d3d-81ed-4a1e-8870-10e94071d2d8/.gitconfig'

5 Temporarily overriding HOME=/home/admin-dev/actions-runner/work/.tmp/d8e75d3d-81ed-4a1e-8870-10e94071d2d8 before making global git config changes

6 Adding repository directory to the temporary git global config as a safe directory

7 /usr/bin/git config --global --add safe.directory /home/admin-dev/actions-runner/work/aceest-web-devops/aceest-web-devops

8 /usr/bin/git config --local --name-only --get-regexp core.sshCommand

9 /usr/bin/git submodule foreach --recursive sh -c "git config --local --name-only --get-regexp 'core.sshCommand' && git config --local --unset-all 'core.sshCommand' || :"

10 /usr/bin/git config --local --name-only --get-regexp http.https://github.com/v\extraheader

11 http.https://github.com/v\extraheader

12 /usr/bin/git config --local --unset-all http.https://github.com/v\extraheader

13 /usr/bin/git submodule foreach --recursive sh -c "git config --local --name-only --get-regexp 'http.https://github.com/v\extraheader' && git config --local --unset-all 'http.https://github.com/v\extraheader' || :"

14 /usr/bin/git config --local --name-only --get-regexp includeIf.gitdir:

15 /usr/bin/git submodule foreach --recursive git config --local --show-origin --name-only --get-regexp remote.origin.url

3. SHADOW DEPLOYMENT

a. [Deployment Spec Files \(GitHub Repositories \)](#)

```
admin-dev@dev-linux: ~/actions-runner

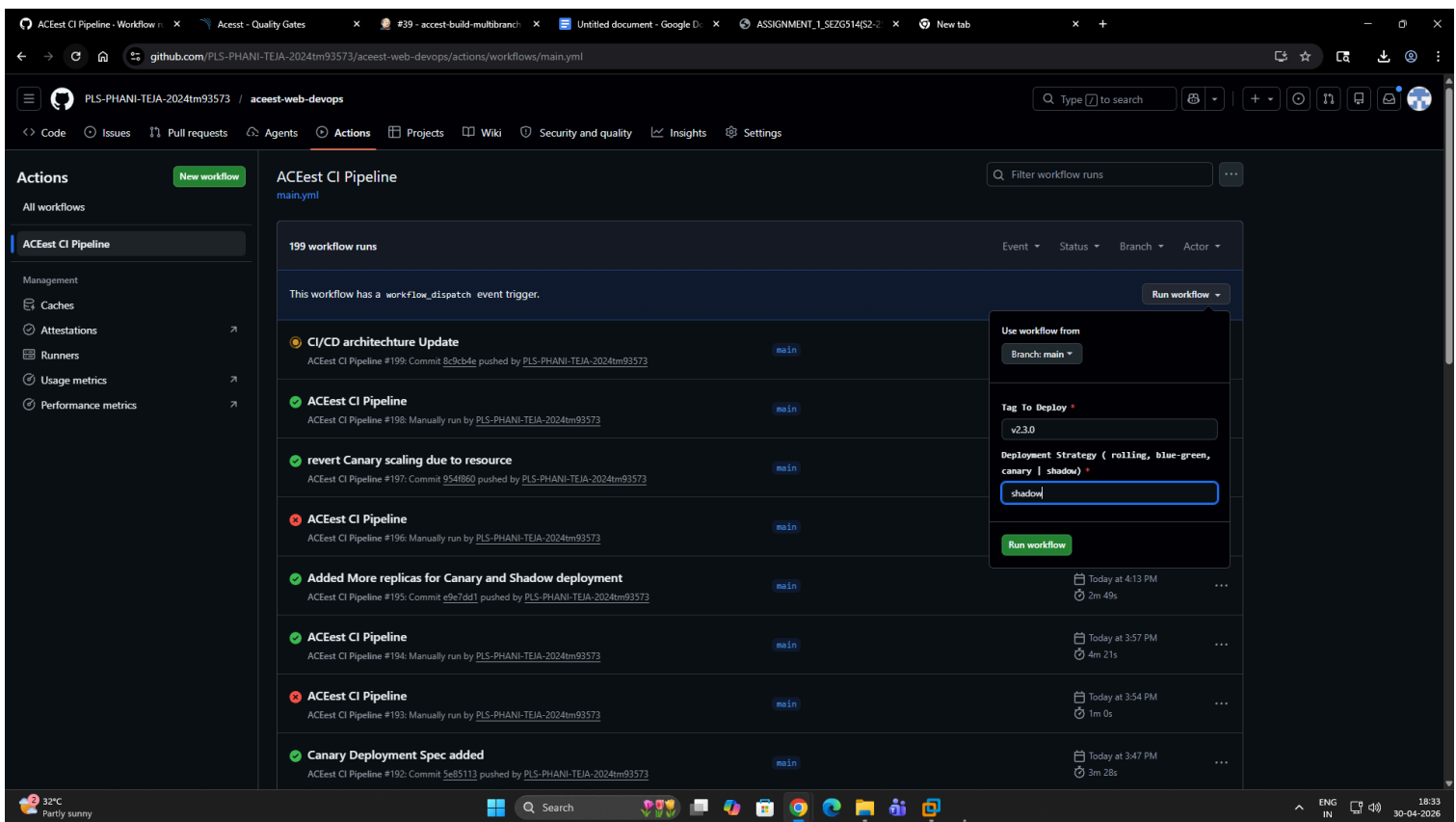
admin-dev@dev-linux:~/actions-runner$ kubectl get pods --show-labels
NAME                                READY   STATUS    RESTARTS   AGE   LABELS
aceest-app-897ccbcfb-64sfc          1/1     Running   0           26m   app=aceest,pod-template-hash=897ccbcfb
aceest-app-897ccbcfb-bhnvt          1/1     Running   0           26m   app=aceest,pod-template-hash=897ccbcfb
aceest-shadow-68cc7594b-59lcx       1/1     Running   0           7m13s app=aceest,pod-template-hash=68cc7594b,track=shadow
aceest-shadow-68cc7594b-6lt5k       1/1     Running   0           7m32s app=aceest,pod-template-hash=68cc7594b,track=shadow

admin-dev@dev-linux:~/actions-runner$ kubectl get po -A
NAMESPACE      NAME                                READY   STATUS    RESTARTS   AGE
default        aceest-app-897ccbcfb-64sfc          1/1     Running   0           26m
default        aceest-app-897ccbcfb-bhnvt          1/1     Running   0           26m
default        aceest-shadow-68cc7594b-59lcx       1/1     Running   0           7m29s
default        aceest-shadow-68cc7594b-6lt5k       1/1     Running   0           7m48s
ingress-nginx  ingress-nginx-admission-create-p6dsf 0/1     Completed 0           4m54s
ingress-nginx  ingress-nginx-admission-patch-zzht8   0/1     Completed 0           4m54s
ingress-nginx  ingress-nginx-controller-596f8778bc-lrkv4 0/1     Running   4 (33s ago) 4m54s
kube-system    coredns-7d76466f9-dgkdh             1/1     Running   3 (114m ago) 19h
kube-system    etcd-minikube                        1/1     Running   3 (114m ago) 19h
kube-system    kube-apiserver-minikube              1/1     Running   3 (114m ago) 19h
kube-system    kube-controller-manager-minikube     1/1     Running   4 (114m ago) 19h
kube-system    kube-proxy-wmfz8                     1/1     Running   3 (114m ago) 19h
kube-system    kube-scheduler-minikube              1/1     Running   3 (114m ago) 19h
kube-system    storage-provisioner                  1/1     Running   7 (113m ago) 19h
kubernetes-dashboard dashboard-metrics-scraper-5565989548-qgpkq 1/1     Running   3 (114m ago) 16h
kubernetes-dashboard kubernetes-dashboard-b84665fb8-bbckw 1/1     Running   6 (113m ago) 16h

admin-dev@dev-linux:~/actions-runner$ kubectl get service
NAME      TYPE      CLUSTER-IP      EXTERNAL-IP      PORT(S)      AGE
aceest-service NodePort  10.96.51.21      <none>           80:30007/TCP 29m
aceest-shadow-service ClusterIP 10.105.18.205    <none>         80/TCP        10m
kubernetes ClusterIP 10.96.0.1         <none>         443/TCP       19h

admin-dev@dev-linux:~/actions-runner$ kubectl get deployments
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
aceest-app 2/2      2             2           29m
aceest-shadow 2/2      2             2           10m

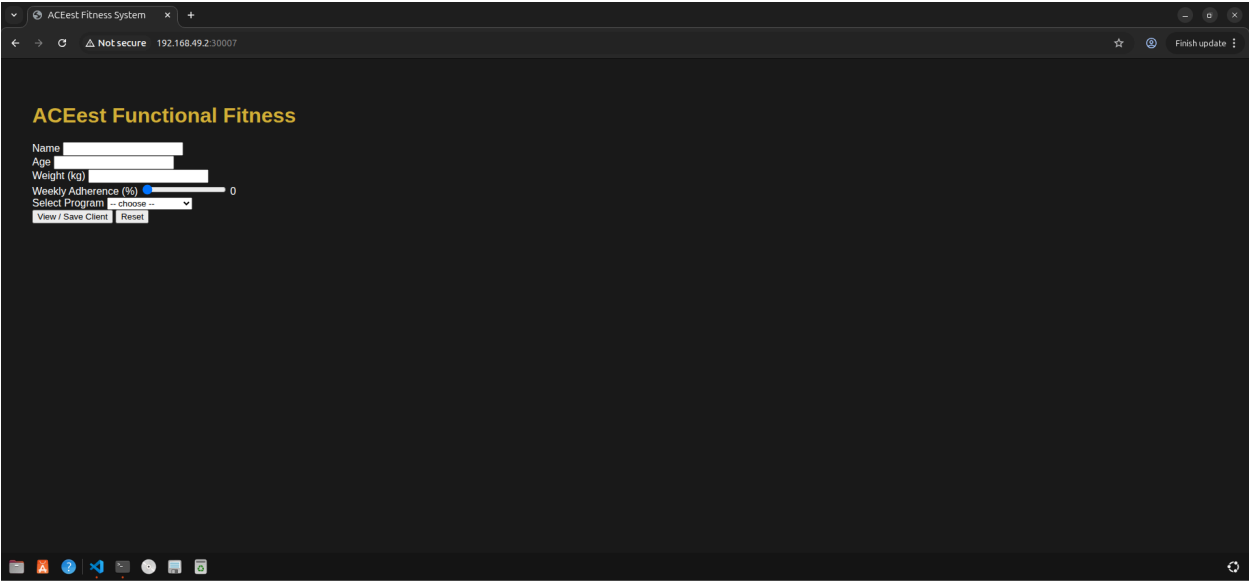
admin-dev@dev-linux:~/actions-runners$
```



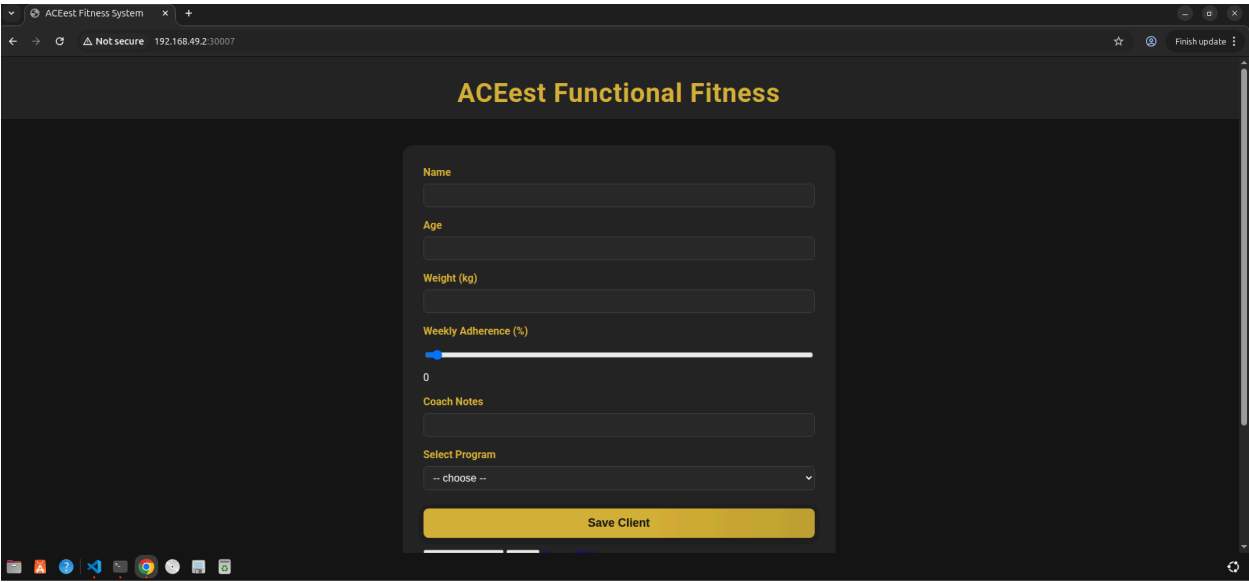
PYTEST CASES (AUTOMATION TEST CASES)
These are downloadable artifacts that can be downloaded via pipeline. [Automation test Cases](#)

APPLICATION UI

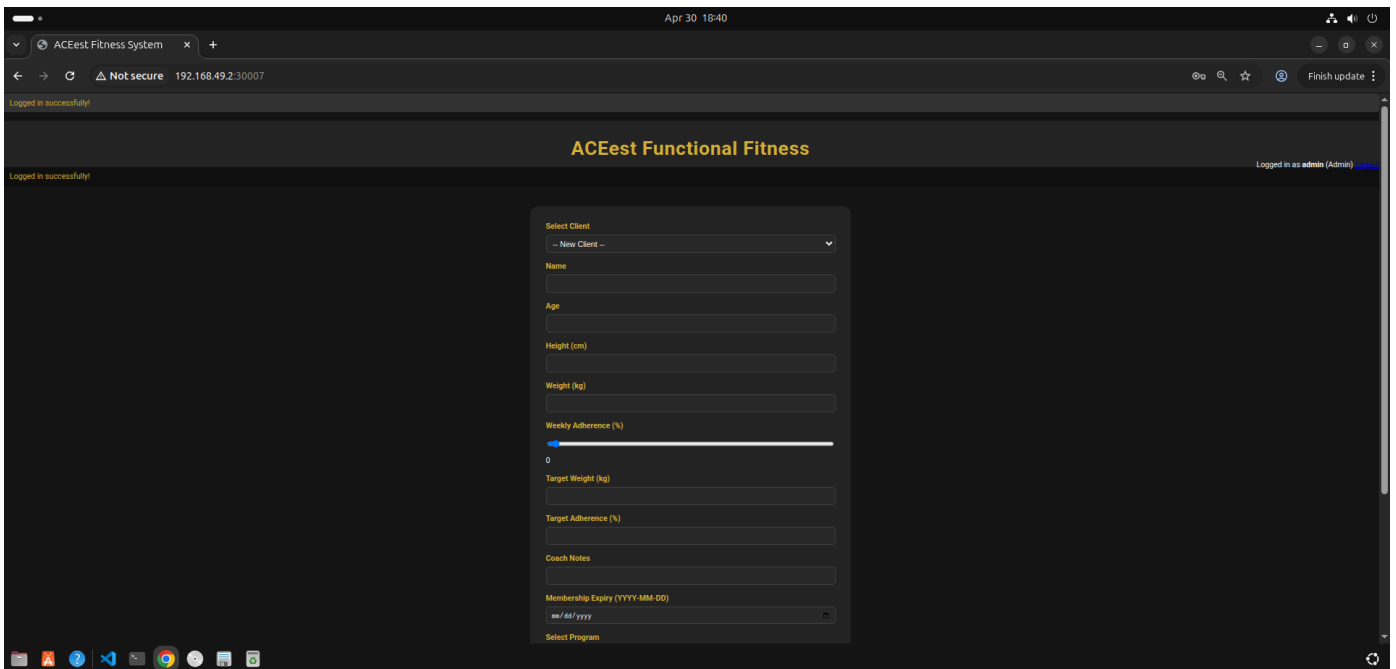
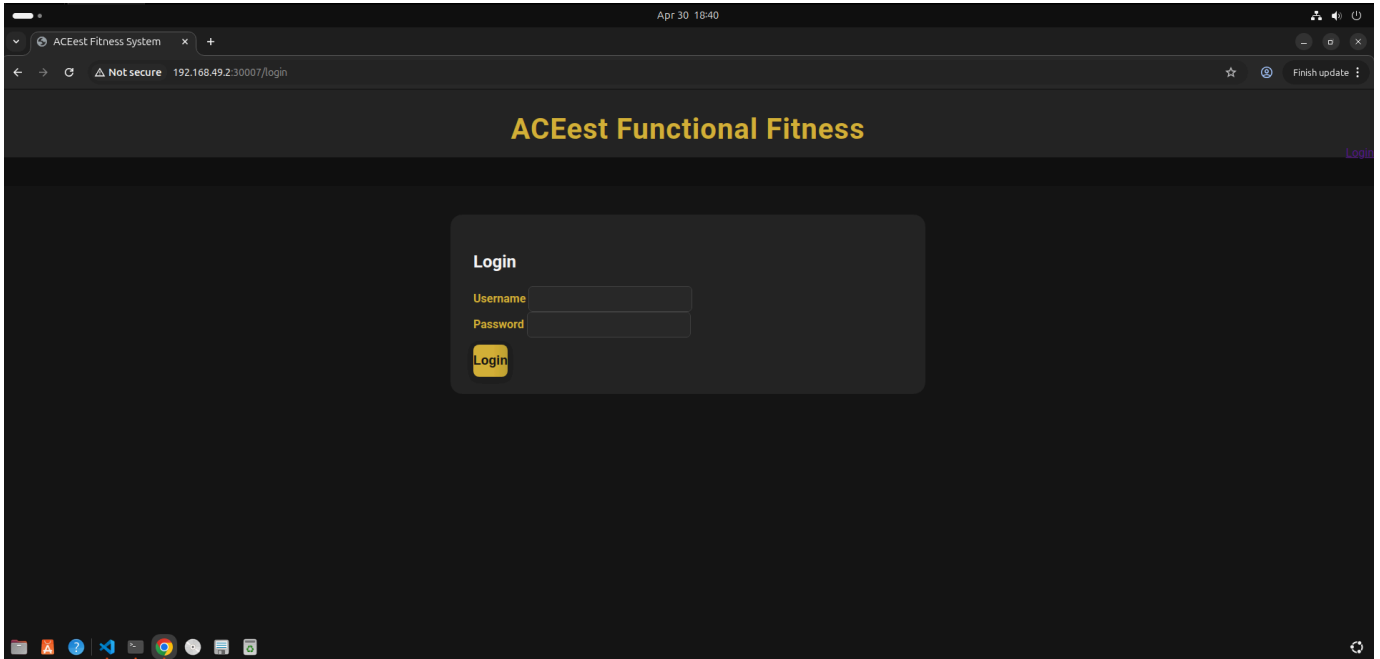
V1.1



V2.1



V3.2.0



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